

```
!pip install gensim
import gensim.downloader as api
import numpy as np
from sklearn.metrics.pairwise import cosine_similarity
from sklearn.decomposition import PCA
import matplotlib.pyplot as plt
```

Collecting gensim

Downloading gensim-4.4.0-cp312-cp312-manylinux_2_24_x86_64.manylinux_2_28_x86_64.whl.metadata
 Requirement already satisfied: numpy>=1.18.5 in /usr/local/lib/python3.12/dist-packages (from gensim==4.4.0)
 Requirement already satisfied: scipy>=1.7.0 in /usr/local/lib/python3.12/dist-packages (from gensim==4.4.0)
 Requirement already satisfied: smart_open>=1.8.1 in /usr/local/lib/python3.12/dist-packages (from gensim==4.4.0)
 Requirement already satisfied: wrapt in /usr/local/lib/python3.12/dist-packages (from gensim==4.4.0)
 Downloading gensim-4.4.0-cp312-cp312-manylinux_2_24_x86_64.manylinux_2_28_x86_64.whl (27.9 MB)
 27.9/27.9 MB 65.1 MB/s eta 0:00:00

Installing collected packages: gensim
 Successfully installed gensim-4.4.0

```
model = api.load("word2vec-google-news-300")
print("Vocabulary size:", len(model.key_to_index))

print("Vector for 'king':")
print(model["king"])
```

```
1.23000000e-01  2.03200125e-01  1.23000000e-01  3.12220000e-02
-1.77734375e-01  8.59375000e-02 -2.18505859e-02  2.05078125e-02
-1.39648438e-01  2.51464844e-02  1.38671875e-01 -1.05468750e-01
 1.38671875e-01  8.88671875e-02 -7.51953125e-02 -2.13623047e-02
 1.72851562e-01  4.63867188e-02 -2.65625000e-01  8.91113281e-03
 1.49414062e-01  3.78417969e-02  2.38281250e-01 -1.24511719e-01
-2.17773438e-01 -1.81640625e-01  2.97851562e-02  5.71289062e-02
-2.89306641e-02  1.24511719e-02  9.66796875e-02 -2.31445312e-01
 5.81054688e-02  6.68945312e-02  7.08007812e-02 -3.08593750e-01
-2.14843750e-01  1.45507812e-01 -4.27734375e-01 -9.39941406e-03
 1.54296875e-01 -7.66601562e-02  2.89062500e-01  2.77343750e-01
-4.86373901e-04 -1.36718750e-01  3.24218750e-01 -2.46093750e-01
-3.03649902e-03 -2.11914062e-01  1.25000000e-01  2.69531250e-01
```

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7.37304688e-02 -9.22851562e-02 1.07910156e-01 1.58203125e-01
4.24804688e-02 1.26953125e-01 3.61328125e-02 2.67578125e-01
-1.01074219e-01 -3.02734375e-01 -5.76171875e-02 5.05371094e-02
5.26428223e-04 -2.07031250e-01 -1.38671875e-01 -8.97216797e-03
-2.78320312e-02 -1.41601562e-01 2.07031250e-01 -1.58203125e-01
1.27929688e-01 1.49414062e-01 -2.24609375e-02 -8.44726562e-02
1.22558594e-01 2.15820312e-01 -2.13867188e-01 -3.12500000e-01
-3.73046875e-01 4.08935547e-03 1.07421875e-01 1.06933594e-01
7.32421875e-02 8.97216797e-03 -3.88183594e-02 -1.29882812e-01
1.49414062e-01 -2.14843750e-01 -1.83868408e-03 9.91210938e-02
1.57226562e-01 -1.14257812e-01 -2.05078125e-01 9.91210938e-02
3.69140625e-01 -1.97265625e-01 3.54003906e-02 1.09375000e-01
1.31835938e-01 1.66992188e-01 2.35351562e-01 1.04980469e-01
-4.96093750e-01 -1.64062500e-01 -1.56250000e-01 -5.22460938e-02
1.03027344e-01 2.43164062e-01 -1.88476562e-01 5.07812500e-02
-9.37500000e-02 -6.68945312e-02 2.27050781e-02 7.61718750e-02
2.89062500e-01 3.10546875e-01 -5.37109375e-02 2.28515625e-01
2.51464844e-02 6.78710938e-02 -1.21093750e-01 -2.15820312e-01
-2.73437500e-01 -3.07617188e-02 -3.37890625e-01 1.53320312e-01
2.33398438e-01 -2.08007812e-01 3.73046875e-01 8.20312500e-02
2.51953125e-01 -7.61718750e-02 -4.66308594e-02 -2.23388672e-02
2.99072266e-02 -5.93261719e-02 -4.66918945e-03 -2.44140625e-01
-2.09960938e-01 -2.87109375e-01 -4.54101562e-02 -1.77734375e-01
-2.79296875e-01 -8.59375000e-02 9.13085938e-02 2.51953125e-01]

```

```

pairs = [
    ("doctor", "nurse"),
    ("cat", "dog"),
    ("car", "bus"),
    ("teacher", "student"),
    ("king", "queen"),
    ("india", "china"),
    ("happy", "joy"),
    ("sad", "cry"),
    ("computer", "laptop"),
    ("school", "college")
]

for w1, w2 in pairs:
    print(f"{w1} - {w2}:", model.similarity(w1, w2))

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doctor - nurse: 0.63195235
cat - dog: 0.76094574
car - bus: 0.4693371
teacher - student: 0.63013655
king - queen: 0.6510957
india - china: 0.35330796
happy - joy: 0.35500747
sad - cry: 0.28272176
computer - laptop: 0.66404927
school - college: 0.6081898

```

```

words = ["king", "university", "football", "hospital", "technology"]

for word in words:
    print(f"\nTop similar words for '{word}':")
    print(model.most_similar(word, topn=5))

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Top similar words for 'king':
[('kings', 0.7138045430183411), ('queen', 0.6510956883430481), ('monarch', 0.6413194537162

Top similar words for 'university':
[('universities', 0.7003918886184692), ('faculty', 0.6780907511711121), ('unversity', 0.67

Top similar words for 'football':
[('soccer', 0.731354832649231), ('fooball', 0.7139959335327148), ('Football', 0.7124834060

Top similar words for 'hospital':
[('Hospital', 0.7931912541389465), ('hopsital', 0.7784007787704468), ('hosptial', 0.758177

Top similar words for 'technology':
[('technologies', 0.8332264423370361), ('innovations', 0.6230790615081787), ('technologica

```

```

print("king - man + woman:")
print(model.most_similar(positive=["king", "woman"], negative=["man"], topn=1))

print("paris - france + india:")
print(model.most_similar(positive=["paris", "india"], negative=["france"], topn=1))

print("teacher - school + hospital:")
print(model.most_similar(positive=["teacher", "hospital"], negative=["school"], topn=1))

```

```

king - man + woman:
[('queen', 0.7118193507194519)]
paris - france + india:
[('chennai', 0.5442505478858948)]
teacher - school + hospital:
[('Hospital', 0.6331106424331665)]

```

```

words = ["king", "queen", "man", "woman",
         "doctor", "nurse", "hospital",
         "school", "teacher", "student",
         "football", "cricket", "tennis",
         "happy", "sad", "joy"]

vectors = np.array([model[w] for w in words])

pca = PCA(n_components=2)
result = pca.fit_transform(vectors)

plt.figure()
plt.scatter(result[:, 0], result[:, 1])

for i, word in enumerate(words):
    plt.annotate(word, xy=(result[i, 0], result[i, 1]))

plt.title("Word Embedding Visualization")
plt.show()

```

